Software Engineering
UA COLLEGE OF ENGINEERING

SFWE 407-507. Foundations of Software Engineering

DR. DIANA SALDANA JIMENEZ

THE UNIVERSITY OF ARIZONA

What is Software?

THE UNIVERSITY OF ARIZONA

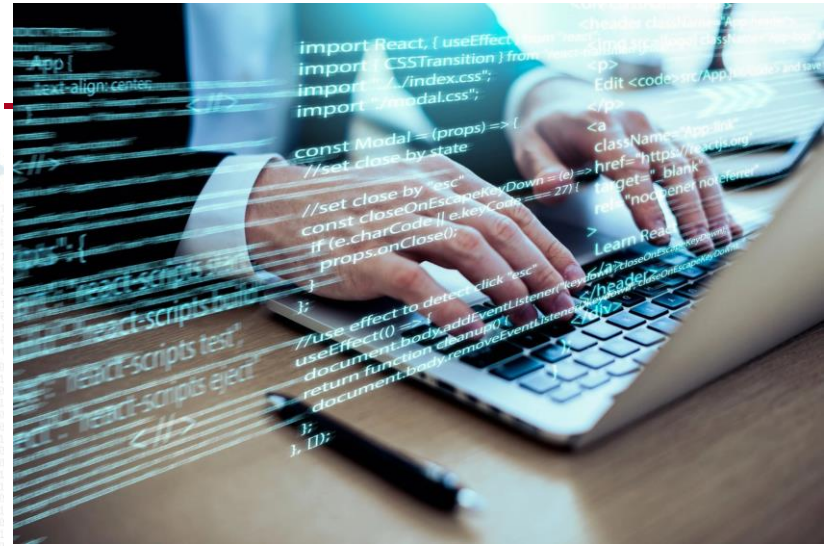
What is Software?

Software Engineering
UA COLLEGE OF ENGINEERING

What is Software?

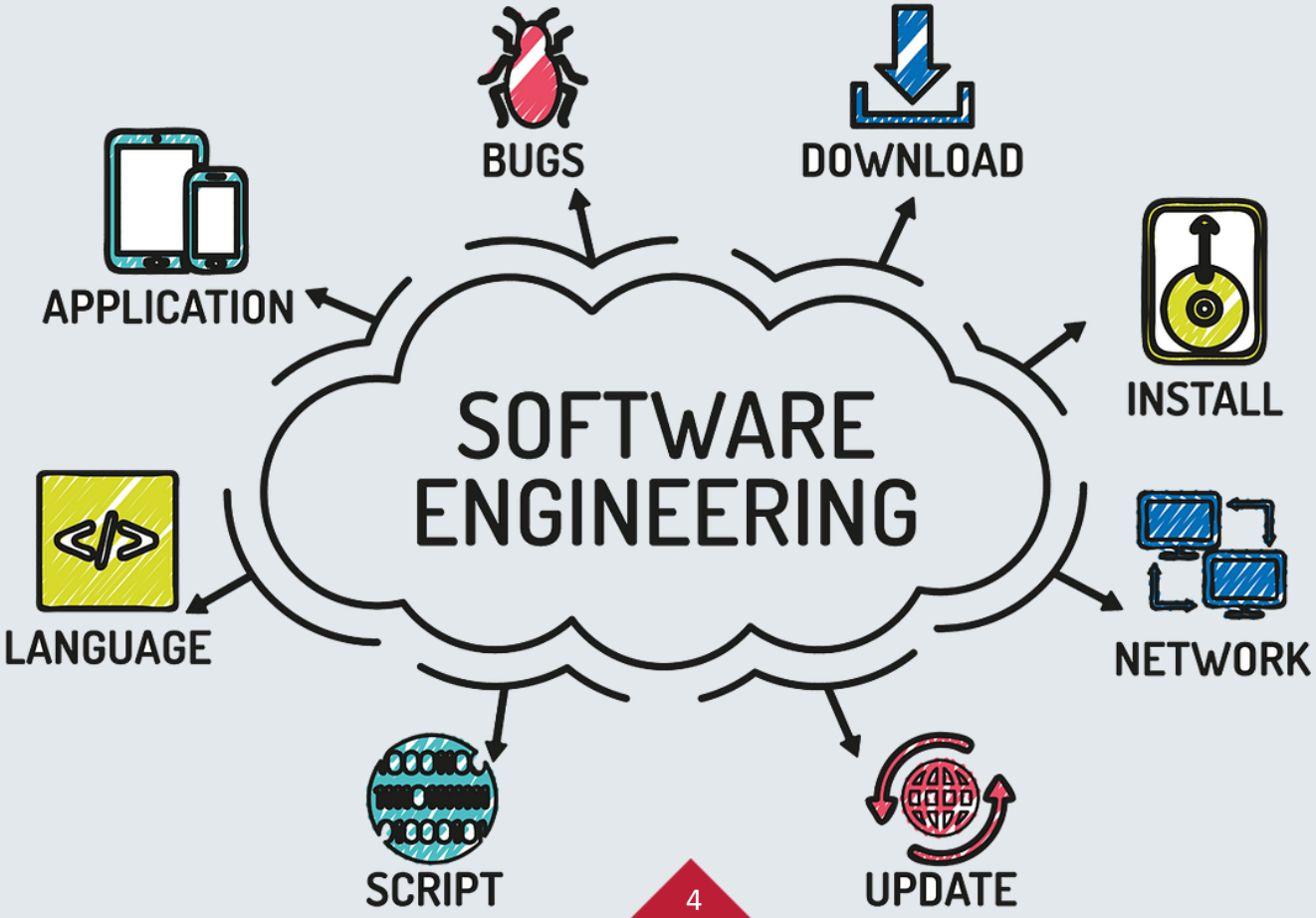
Software is a set of computer programs that instruct computers what to do. It mainly comprises code, algorithms, and data structures necessary to operate a computer system.

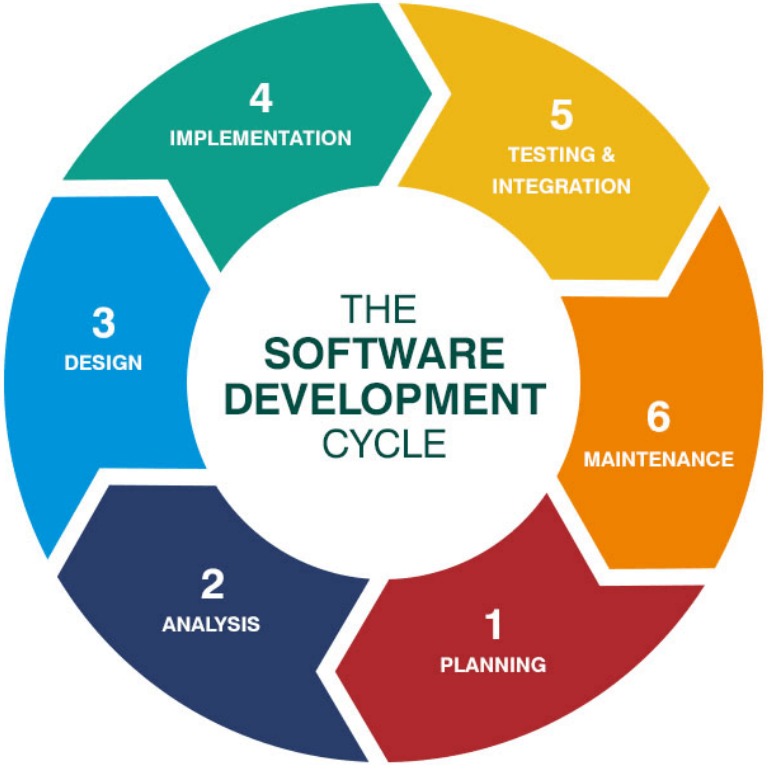
FINOIT



THE UNIVERSITY OF ARIZONA

What is Software Engineering?





Why Software Engineering?

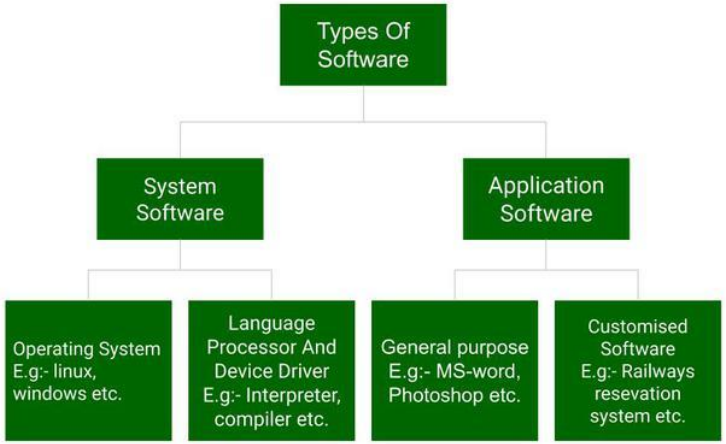
Software engineering is intended to support professional software development rather than individual programming. It includes techniques that support program specification, design, and evolution, none of which are normally relevant for personal software development.

Attributes of Good Software

Product characteristic	Description
Acceptability	Software must be acceptable to the type of users for which it is designed. This means that it must be understandable, usable, and compatible with other systems that they use.
Dependability and security	Software dependability includes a range of characteristics including reliability, security, and safety. Dependable software should not cause physical or economic damage in the event of system failure. Software has to be secure so that malicious users cannot access or damage the system.
Efficiency	Software should not make wasteful use of system resources such as memory and processor cycles. Efficiency therefore includes responsiveness, processing time, resource utilization, etc.
Maintainability	Software should be written in such a way that it can evolve to meet the changing needs of customers. This is a critical attribute because software change is an inevitable requirement of a changing business environment.

Source: Software Engineering, Ian Sommerville.

Types of Software Systems



Software Engineering Process Activities

- **Software Specification,** where customers and engineers define the software that is to be produced and the constraints on its operation.
- **Software Development,** where the software is designed and programmed.
- **Software Validation,** where the software is checked to ensure that it is what the customer requires
- **Software Evolution,** where the software is modified to reflect changing customer and market requirements.

Application Types

- Stand-alone Applications
  - Interactive Transaction-based Applications
  - Embedded Control Systems
  - Batch Processing Systems
  - Entertainment Systems
  - Modeling and Simulation Systems
  - Data Collection Systems
  - Systems of Systems

Software Engineering Fundamentals

- Development Process
- Dependability and Performance
- Managing Software Specification and Requirements
- Software Reuse

Web-based Software Engineering

- Complex Distributed Systems
- Web Services
- Cloud Computing
- Software Reuse

Software Engineering Ethics

THE UNIVERSITY OF ARIZONA

Association for Computing Machinery's (ACM) Code of Ethics

ACM Ethics

The Official Site of the Association for Computing Machinery's Committee on Professional Ethics

acm Committee on Professional Ethics

About Code of Ethics Enforcement Integrity Project

This Code was developed by the ACM/IEEE-CS joint task force on Software Engineering Ethics and Professional Practices (SEEPP):

Executive Committee: Donald Gotterbarn (Chair), Keith Miller and Simon Rogerson;

Members: Steve Barber, Peter Barnes, Ilene Burnstein, Michael Davis, Amr El-Kadi, N. Ben Fairweather, Milton Fulghum, N. Jayaram, Tom Jewett, Mark Kanko, Ernie Kallman, Duncan Langford, Joyce Currie Little, Ed Mechler, Manuel J. Norman, Douglas Phillips, Peter Ron Prinzivalli, Patrick Sullivan, John Weckert, Vivian Weil, S. Weisband and Laurie Honour Werth.

This Code may be published without permission as long as it is not changed in any way and it carries the copyright notice. Copyright (c) 1999 by the Association for Computing Machinery, Inc. and the Institute for Electrical and Electronics Engineers, Inc.

ACM / IEEE-CS Joint Task Force has developed a Software Engineering Code of Ethics in 1997 https://ethics.acm.org/code-of-ethics/software-engineering-code/

Consists of 8 core principles, each having detailed description of lower-level attributes

Principle 1: PUBLIC

Software Engineers shall act consistently with the public interest.

1.01. Accept full responsibility for their own work.

1.02. Moderate the interests of the software engineer, the employer, the client and the users with the public good.

1.03. Approve software only if they have a well-founded belief that it is safe, meets specifications, passes appropriate tests, and does not diminish quality of life, diminish privacy or harm the environment. The ultimate effect of the work should be to the public good.

1.04. Disclose to appropriate persons or authorities any actual or potential danger to the user, the public, or the environment, that they reasonably believe to be associated with software or related documents.

1.05. Cooperate in efforts to address matters of grave public concern caused by software, its installation, maintenance, support or documentation.

1.06. Be fair and avoid deception in all statements, particularly public ones, concerning software or related documents, methods and tools.

1.07. Consider issues of physical disabilities, allocation of resources, economic disadvantage and other factors that can diminish access to the benefits of software.

1.08. Be encouraged to volunteer professional skills to good causes and contribute to public education concerning the discipline.

Principle 2: CLIENT AND EMPLOYER

Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.

2.01. Provide service in their areas of competence, being honest and forthright about any limitations of their experience and education.

2.02. Not knowingly use software that is obtained or retained either illegally or unethically.

2.03. Use the property of a client or employer only in ways properly authorized, and with the client's or employer's knowledge and consent.

2.04. Ensure that any document upon which they rely has been approved, when required, by someone authorized to approve it.

2.05. Keep private any confidential information gained in their professional work, where such confidentiality is consistent with the public interest and consistent with the law.

2.06. Identify, document, collect evidence and report to the client or the employer promptly if, in their opinion, a project is likely to fail, to prove too expensive, to violate intellectual property law, or otherwise to be problematic.

2.07. Identify, document, and report significant issues of social concern, of which they are aware, in software or related documents, to the employer or the client.

2.08. Accept no outside work detrimental to the work they perform for their primary employer.

2.09. Promote no interest adverse to their employer or client, unless a higher ethical concern is being compromised; in that case, inform the employer or another appropriate authority of the ethical concern.

Principle 3: PRODUCT

Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.

3.01. Strive for high quality, acceptable cost and a reasonable schedule, ensuring significant tradeoffs are clear to and accepted by the employer and the client, and are available for consideration by the user and the public.

3.02. Ensure proper and achievable goals and objectives for any project on which they work or propose.

3.03. Identify, define and address ethical, economic, cultural, legal and environmental issues related to work projects.

3.04. Ensure that they are qualified for any project on which they work or propose to work by an appropriate combination of education and training, and experience.

3.05. Ensure an appropriate method is used for any project on which they work or propose to work.

3.06. Work to follow professional standards, when available, that are most appropriate for the task at hand, departing from these only when ethically or technically justified.

3.07. Strive to fully understand the specifications for software on which they work.

3.08. Ensure that specifications for software on which they work have been well documented, satisfy the users' requirements and have the appropriate approvals.

3.09. Ensure realistic quantitative estimates of cost, scheduling, personnel, quality and outcomes on any project on which they work or propose to work and provide an uncertainty assessment of these estimates.

Principle 3 *(Continued):* PRODUCT

Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.

3.10. Ensure adequate testing, debugging, and review of software and related documents on which they work.

3.11. Ensure adequate documentation, including significant problems discovered and solutions adopted, for any project on which they work.

3.12. Work to develop software and related documents that respect the privacy of those who will be affected by that software.

3.13. Be careful to use only accurate data derived by ethical and lawful means, and use it only in ways properly authorized.

3.14. Maintain the integrity of data, being sensitive to outdated or flawed occurrences.

3.15. Treat all forms of software maintenance with the same professionalism as new development.

Principle 4: JUDGEMENT

Software engineers shall maintain integrity and independence in their professional judgement.

4.01. Temper all technical judgments by the need to support and maintain human values.

4.02. Only endorse documents either prepared under their supervision or within their areas of competence and with which they are in agreement.

4.03. Maintain professional objectivity with respect to any software or related documents they are asked to evaluate.

4.04. Not engage in deceptive financial practices such as bribery, double billing, or other improper financial practices.

4.05. Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped.

4.06. Refuse to participate, as members or advisors, in a private, governmental or professional body concerned with software related issues, in which they, their employers or their clients have undisclosed potential conflicts of interest.

Let’s Practice….

THE UNIVERSITY OF ARIZONA

What do you think?

Software Engineering
UA COLLEGE OF ENGINEERING

➢ SELECT THE BEST ANSWER

- **Scenario:** You are asked to sign off on a project report that contains data that was slightly manipulated to make results appear more favorable. What should you do?
- A) Sign the report as is, as it is just a minor discrepancy.
- B) Refuse to sign the report and explain your concerns to your supervisor.
- C) Sign the report but include a disclaimer noting your reservations.
- D) alter the data further to make it more aligned with expectations.
- E) or do you have other choice?

THE UNIVERSITY OF ARIZONA

What do you think?

Software Engineering
UA COLLEGE OF ENGINEERING

➢ SELECT THE BEST ANSWER

- **Scenario:** A client asks you to design a product that could potentially harm the environment if not used properly. What is the most ethical approach?

- A) Design the product as requested but provide clear warnings and instructions on proper usage.
- B) Refuse to design the product due to potential environmental harm.
- C) Design a modified version of the product that minimizes environmental impact.
- D) Suggest the client consider alternative products that are less environmentally damaging.
- E) or do you have other choice?

THE UNIVERSITY OF ARIZONA

Principle 5: MANAGEMENT

Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.

5.01. Ensure good management for any project on which they work, including effective procedures for promotion of quality and reduction of risk.

5.02. Ensure that software engineers are informed of standards before being held to them.

5.03. Ensure that software engineers know the employer's policies and procedures for protecting passwords, files and information that is confidential to the employer or confidential to others.

5.04. Assign work only after taking into account appropriate contributions of education and experience tempered with a desire to further that education and experience.

5.05. Ensure realistic quantitative estimates of cost, scheduling, personnel, quality and outcomes on any project on which they work or propose to work, and provide an uncertainty assessment of these estimates.

5.06. Attract potential software engineers only by full and accurate description of the conditions of employment.

5.07. Offer fair and just remuneration.

5.08. Not unjustly prevent someone from taking a position for which that person is suitably qualified.

5.09. Ensure that there is a fair agreement concerning ownership of any software, processes, research, writing, or other intellectual property to which a software engineer has contributed.

5.10. Provide for due process in hearing charges of violation of an employer's policy or of this Code.

5.11. Not ask a software engineer to do anything inconsistent with this Code.

5.12. Not punish anyone for expressing ethical concerns about a project.

Principle 6: PROFESSION

Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

6.01. Help develop an organizational environment favorable to acting ethically.

6.02. Promote public knowledge of software engineering.

6.03. Extend software engineering knowledge by appropriate participation in professional organizations, meetings and publications.

6.04. Support, as members of a profession, other software engineers striving to follow this Code.

6.05. Not promote their own interest at the expense of the profession, client or employer.

6.06. Obey all laws governing their work, unless, in exceptional circumstances, such compliance is inconsistent with the public interest.

6.07. Be accurate in stating the characteristics of software on which they work, avoiding not only false claims but also claims that might reasonably be supposed to be speculative, vacuous, deceptive, misleading, or doubtful.

6.08. Take responsibility for detecting, correcting, and reporting errors in software and associated documents on which they work.

6.09. Ensure that clients, employers, and supervisors know of the software engineer's commitment to this Code of Ethics, and the subsequent ramifications of such commitment.

Principle 6 *(Continued):* PROFESSION

Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

6.10. Avoid associations with businesses and organizations which are in conflict with this code.

6.11. Recognize that violations of this Code are inconsistent with being a professional software engineer.

6.12. Express concerns to the people involved when significant violations of this Code are detected unless this is impossible, counter-productive, or dangerous.

6.13. Report significant violations of this Code to appropriate authorities when it is clear that consultation with people involved in these significant violations is impossible, counter-productive or dangerous.

Principle 7: COLLEAGUES

Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

7.01. Encourage colleagues to adhere to this Code.

7.02. Assist colleagues in professional development.

7.03. Credit fully the work of others and refrain from taking undue credit.

7.04. Review the work of others in an objective, candid, and properly-documented way.

7.05. Give a fair hearing to the opinions, concerns, or complaints of a colleague.

7.06. Assist colleagues in being fully aware of current standard work practices including policies and procedures for protecting passwords, files and other confidential information and security measures in general.

7.07. Not unfairly intervene in the career of any colleague; however, concern for the employer, the client or public interest may compel software engineers, in good faith, to question the competence of a colleague.

7.08. In situations outside of their own areas of competence, call upon the opinions of other professionals who have competence in that area.

8.01. Further their knowledge of developments in the analysis, specification, design, development, maintenance and testing of software and related documents, together with the management of the development process.

8.02. Improve their ability to create safe, reliable, and useful quality software at reasonable cost and within a reasonable time.

8.03. Improve their ability to produce accurate, informative, and well-written documentation.

8.04. Improve their understanding of the software and related documents on which they work and of the environment in which they will be used.

8.05. Improve their knowledge of relevant standards and the law governing the software and related documents on which they may work.

8.06. Improve their knowledge of this Code, its interpretation, and its application to their work.

8.07. Not give unfair treatment to anyone because of any irrelevant prejudices.

8.08. Not influence others to undertake any action that involves a breach of this Code.

8.09. Recognize that personal violations of this Code are inconsistent with being a professional software engineer.

Principle 8: SELF

Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Let’s Practice….

THE UNIVERSITY OF ARIZONA

What do you think?

Software Engineering
UA COLLEGE OF ENGINEERING

➢ SELECT THE BEST ANSWER

- Scenario: You discover that a colleague is knowingly using substandard materials in a project to cut costs. What is your ethical responsibility?

- A) Do nothing as it is not your direct responsibility.
- B) Privately confront your colleague and urge them to correct the issue.
- C) Immediately report the issue to your supervisor.
- D) ignore the situation and let the consequences play out.
- E) or do you have other choice?

THE UNIVERSITY OF ARIZONA

What do you think?

Software Engineering
UA COLLEGE OF ENGINEERING

➢ SELECT THE BEST ANSWER

- A company offers you a large bonus to expedite a project, potentially compromising safety standards. What should you do?

- A) Accept the bonus and prioritize the project timeline.
- B) Negotiate the bonus terms to ensure safety is not compromised.
- C) Decline the bonus and explain your ethical concerns.
- D) Inform your colleagues about the offer and ask for their input.
- E) or do you have other choice?

THE UNIVERSITY OF ARIZONA

Let’s discuss what AI thinks

THE UNIVERSITY OF ARIZONA

The Self-Driving Car Dilemma

- **Scenario:** You're developing the software for a self-driving car. In an unavoidable accident, the car must choose between: Hitting a pedestrian or Swerving and hitting a child playing in the road.
- **Ethical Considerations:** Utilitarianism: Prioritize the greatest good for the greatest number.
- Deontology: Follow moral rules (e.g., "Do not kill").
- Virtue Ethics: Focus on developing moral character (e.g., courage, compassion).
- **Potential Answers (with no easy right answer): Prioritize pedestrians:** Based on the idea that adults are more likely to have lived a full life.
- **Prioritize children:** Based on the idea that children have more potential life ahead of them.
- **Randomize:** Avoid bias and potential discrimination.
- **Human Override:** Allow a human passenger to make the split-second decision.

Data Privacy Violation

- **Scenario:** You discover a bug in your company's social media app that inadvertently collects and stores users' precise location data without their explicit consent. **Ethical Considerations:** Respect for user privacy.
- Data protection laws (e.g., GDPR).
- Company policies and ethical guidelines.
- **Potential Answers: Report the bug immediately:** To your manager, the security team, and potentially an internal ethics board.
- **Document your concerns:** Keep records of your actions and communications.
- **Consider escalating the issue:** If internal efforts are unsuccessful, consider contacting a regulatory body.

Biased Algorithms

Software Engineering
UA COLLEGE OF ENGINEERING

- **Scenario:** You're developing a machine learning model for loan applications. You notice the model consistently denies loans to applicants from certain zip codes, even if their financial profiles are strong. **Ethical Considerations:** Fairness and non-discrimination.
- Potential for algorithmic bias to perpetuate existing societal inequalities.
- Transparency and explainability of AI systems.
- **Potential Answers: Investigate the root cause of the bias:** Analyze the training data for potential biases, check for feature engineering issues, and ensure the model is not relying on irrelevant or discriminatory factors.
- **Re-train the model:** Use more diverse and representative data, implement bias mitigation techniques (e.g., fairness constraints), and regularly monitor the model's performance for signs of bias.
- **Communicate concerns:** Discuss the issue with your team, data scientists, and potentially stakeholders to find a solution.

THE UNIVERSITY OF ARIZONA

Important Notes:

Software Engineering
UA COLLEGE OF ENGINEERING

- These are complex ethical dilemmas with no easy answers.
- The best course of action often depends on the specific context, company culture, and legal requirements.
- It's crucial for software engineers to be aware of ethical guidelines, professional codes of conduct (e.g., ACM Code of Ethics), and relevant laws and regulations.
- Open and honest communication, critical thinking, and a commitment to ethical principles are essential in navigating these challenges.
- **Disclaimer:** These scenarios and potential answers are for illustrative purposes only and do not constitute legal or ethical advice.
- I encourage you to explore these scenarios further and engage in thoughtful discussions with colleagues and mentors to develop your own understanding of ethical decision-making in software engineering.

THE UNIVERSITY OF ARIZONA

Legal Consequences

Software Engineering
UA COLLEGE OF ENGINEERING

- Do you know about other recent case?

The New York Times

Volkswagen Engineer Gets Prison in Diesel Cheating Case

> Mr. Liang, who helped develop the software that concealed high levels of pollutants generated by Volkswagen's diesel engines, reached a plea deal with prosecutors last year after agreeing to assist in the government's investigation of the company.

Former GE engineer gets two-year prison term in espionage case

Xiaoqing Zheng, 59, of Niskayuna faced a judge 10 months after a jury convicted him of one count of conspiring to commit economic espionage.

Robert Gavin
Jan. 3, 2023 | Updated: Jan. 4, 2023 8:19 a.m.

MONEYWATCH ›

Former Theranos CEO Elizabeth Holmes reports to prison to serve her 11-year sentence for fraud

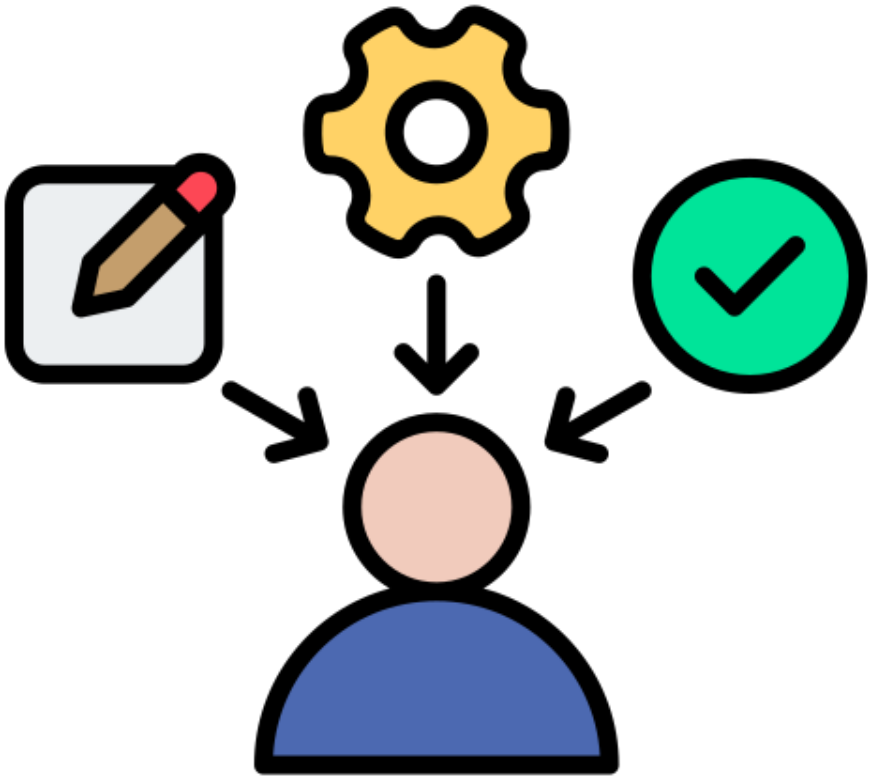
Third ex-Apple engineer charged over Apple Car technology theft

Malcolm Owen | May 16, 2023

THE UNIVERSITY OF ARIZONA

Issues of professional responsibility

- **Confidentiality**, *you should normally respect the confidentiality of your employers or clients regardless of whether or not a formal confidentiality agreement has been signed.*
- **Competence,** *you should not misrepresent your level of competence.*
- **Intellectual property rights,** *you should be aware of local laws governing the use of intellectual property such as patents and copyright.*
- **Computer misuse,** *you should not use your technical skills to misuse other people's computers. e.g. game playing on an employer's machine, dissemination of viruses or other malware.*



Case Studies

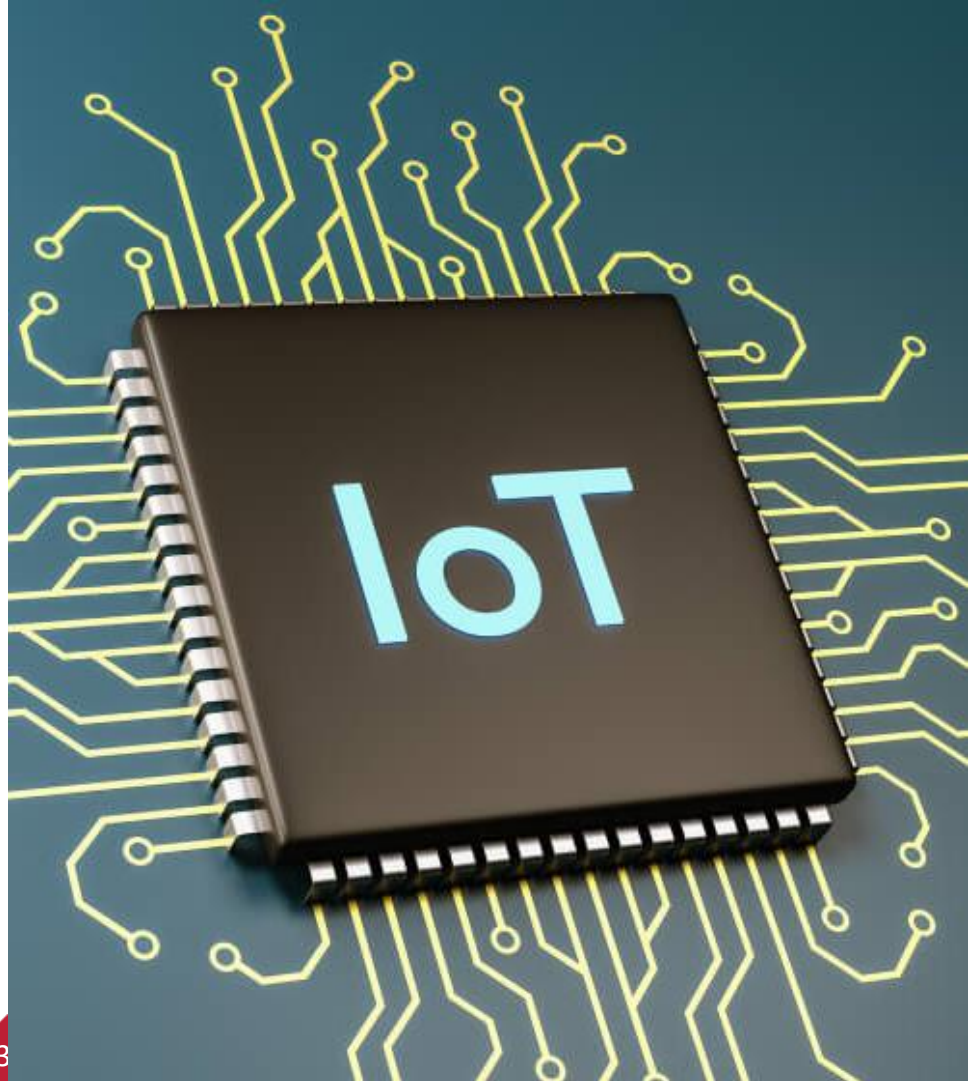
THE UNIVERSITY OF ARIZONA

Why Case Studies?

- Real-World Examples
- Illustrate the challenges
- Exemplify best practices

Embedded System

- Insulin Pump System
  - Controls a hardware device and is embedded in that device.
  - Concerns:
    - Physical size
    - Responsiveness
    - Power Management



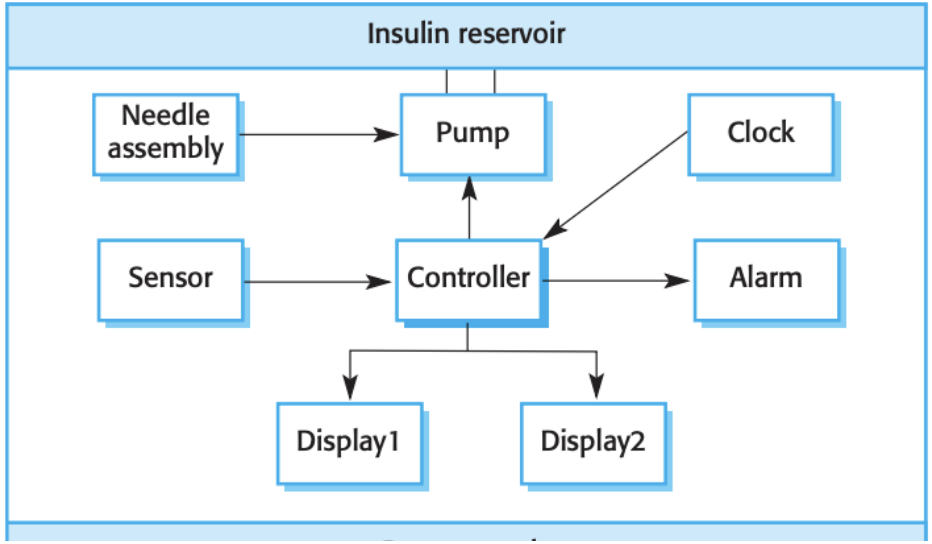
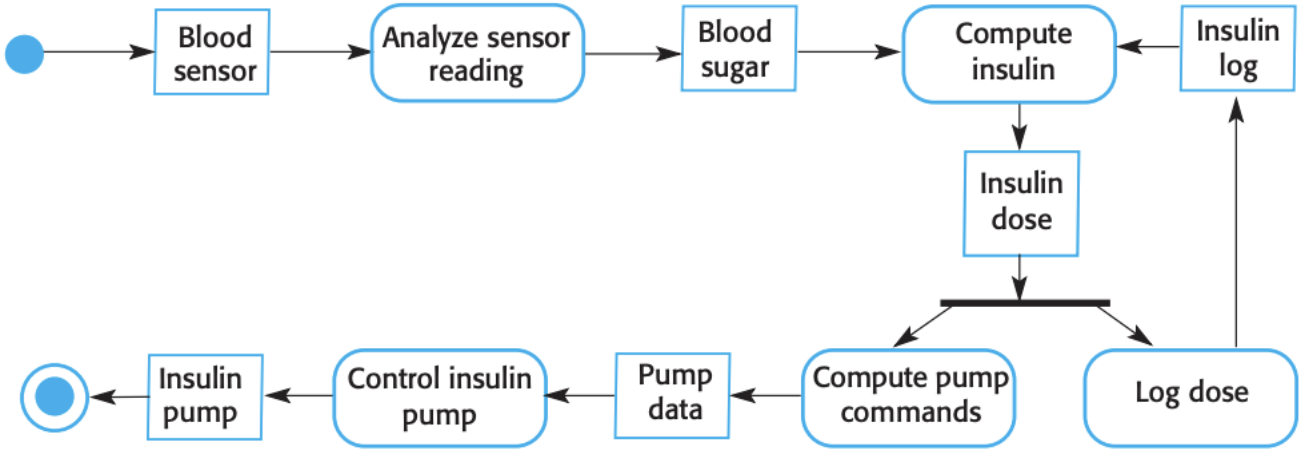


Figure 1.4 Insulin pump hardware architecture

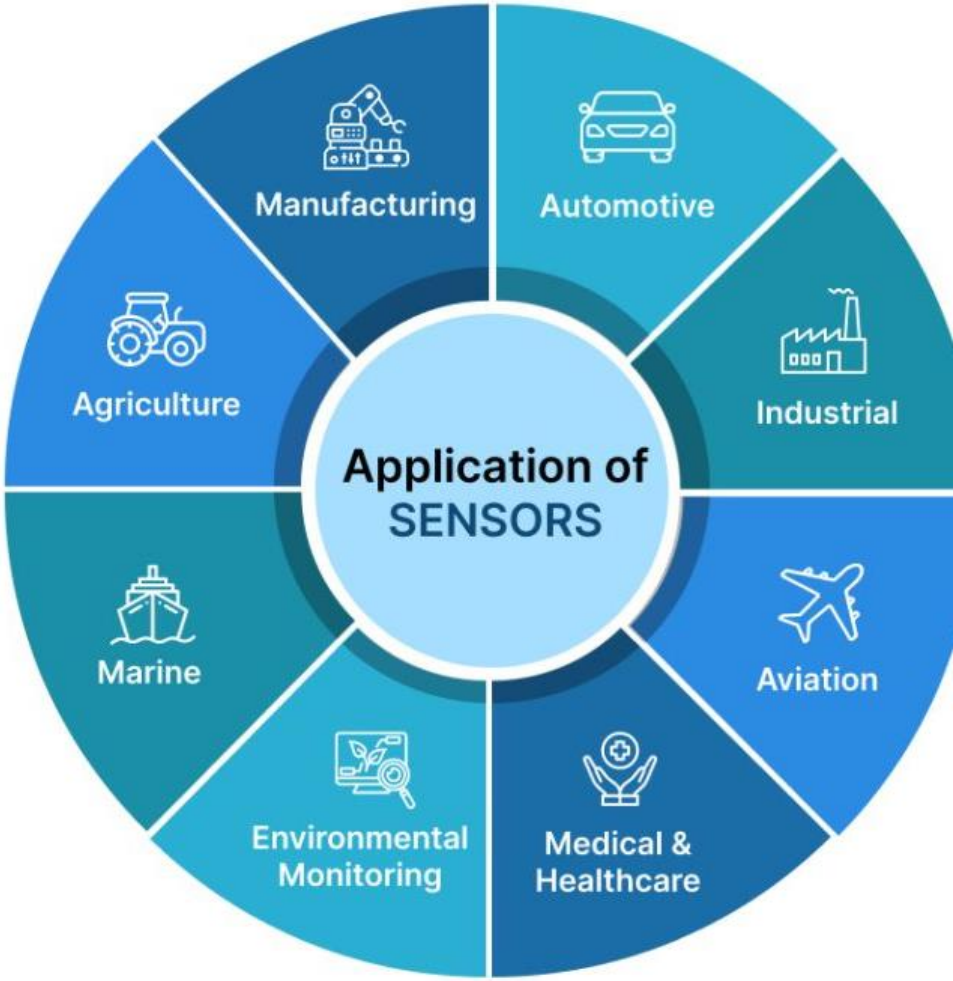


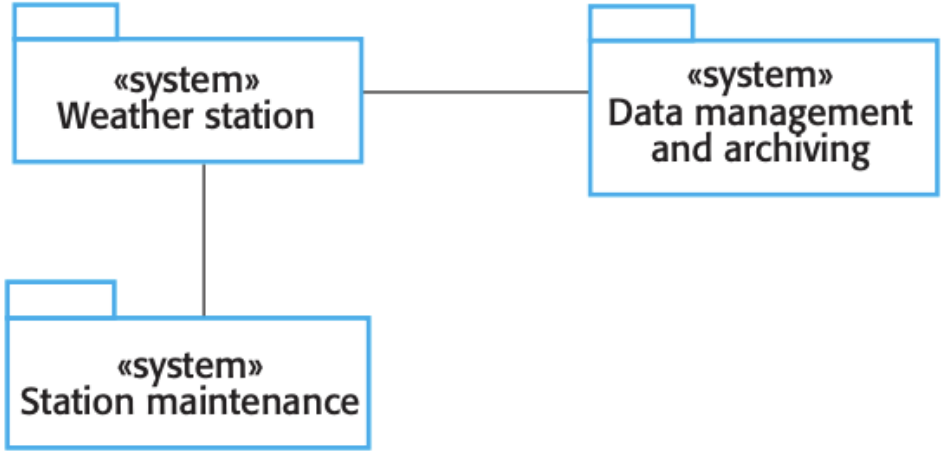
Information Systems

- Medical Records
  - Manages a database of patient information
  - Concerns:
    - Security
    - Usability
    - Privacy
    - Data Integrity

Sensor-based data collection system

- Wilderness Weather Station
  - Collects data form sensors and processes that data.
  - Concerns:
    - Reliability
    - Maintainability




«system»
Weather station
«system»
Data management
and archiving
«system»
Station maintenance

Support System

- Digital Learning System
  - Integrated collection of software tools used to support some kind of activity.
  - Concerns:
    - Domain Specific
    - Governance
    - Compliance

Browser-based user interface | iLearn app

Configuration services

Group management | Application management | Identity management

Application services

Email | Messaging | Video conferencing | Newspaper archive
Word processing | Simulation | Video storage | Resource finder
Spreadsheet | Virtual learning environment | History archive

Utility services

Authentication | Logging and monitoring | Interfacing
User storage | Application storage | Search